
NASH BARGAINING PREFERENCE OPTIMIZATION

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ABSTRACT

Aligning one large language model (LLM) to several objectives requires two decisions: how to represent preferences within each objective and how to choose one compromise across objectives. The first decision is difficult because pairwise judgments can be cyclic in realistic settings. For a medical-advice prompt, annotators may prefer a concise answer to a verbose one, the verbose answer to an incomplete one, yet the incomplete answer to the concise one because it includes a critical safety caveat. No scalar ranking can satisfy all three comparisons. Nash Learning from Human Feedback avoids this restriction by treating pairwise preferences as game payoffs, but its multi-objective extensions still require an aggregation rule. We introduce Nash Bargaining Preference Optimization (NBPO), which evaluates each objective through a regularized preference game and applies Nash bargaining to its improvement over a common reference policy. Our main technical contribution is a practical optimizer for these policy-dependent game values: a constrained KL-proximal update handles zero surplus at the reference, a prompt-separable finite-pool solve accounts for adaptive opponents, and pairwise log-ratio regression transfers the resulting policy target to an LLM without evaluating an intractable normalizer. Under exact population updates, the final iterate achieves convergence with rate $O(1/T)$. Controlled experiments show that adaptive game-valued objectives remain robust as preference nontransitivity increases, while natural multi-objective experiments evaluate Nash bargaining under conflicting alignment objectives.

1 INTRODUCTION

Reinforcement Learning from Human Feedback (RLHF) is a standard approach for aligning large language models (LLMs) with human judgments (Christiano et al., 2017; Ouyang et al., 2022). Standard RLHF learns a scalar reward from pairwise comparisons. Direct Preference Optimization removes the separately trained reward model, but still assumes that comparisons are generated by an implicit scalar reward through a Bradley–Terry (BT) model (Rafailov et al., 2023; Bradley & Terry, 1952). Both approaches therefore require every response to admit one scalar score whose ordering explains all pairwise preferences.

This assumption can fail even within a single alignment objective. Human judgments can reflect several latent attributes whose relative importance changes with the prompt and the pair being compared (Tversky, 1969; Wang et al., 2024). As an illustrative example, consider three answers to a medical-advice question. Response A is concise and actionable but omits an important caveat; response B is comprehensive but verbose; and response C is less complete but includes the critical safety warning. Annotators may prefer A to B for relevance, B to C for completeness, and C to A for safety and reliability. These comparisons are cyclic: $A \succ B$, $B \succ C$, and $C \succ A$. Fitting the first two comparisons with a scalar reward requires $r(A) > r(B) > r(C)$, which contradicts the third. As a result, optimizing the fitted scalar reward can favor a response that appears best under the learned ranking but is still beaten by another response according to the original pairwise judgments (Munos et al., 2024).

Nash Learning from Human Feedback (NLHF) avoids this misspecification by treating prompt-conditioned comparisons directly as payoffs in a zero-sum preference game, and INPO, SPPO, and related methods provide scalable equilibrium-seeking updates (Munos et al., 2024; Zhang et al., 2025; Wu et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Extending this view to multiple objectives introduces a second problem: after representing each objective without imposing transitivity, which compromise should one deployed policy realize? Scalar multi-

objective methods expose this trade-off through weights, priorities, constraints, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b; Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). Recent game-theoretic methods preserve general pairwise feedback but combine objectives through Blackwell target sets or a worst-objective criterion (Blackwell, 1956; Bhatia et al., 2020; Zhang et al., 2026).

Classical Nash bargaining offers a priority-free way to select one compromise. It maximizes the product of each objective’s improvement over a common fallback (Nash, 1950). In alignment, the reference policy provides a natural fallback: an aligned policy should improve each objective relative to the model that would otherwise be deployed. The classical Nash solution is Pareto optimal and invariant to independent changes in the units used for different objectives. Prior alignment work has applied Nash welfare to learned scalar rewards, while game-based preference methods have focused mainly on minimax or Blackwell solutions (Zhong et al., 2024a; Navon et al., 2022; Zhang et al., 2026). What remains is a practical method for Nash bargaining when every objective value is itself defined by a policy-dependent preference game rather than by a fixed scalar reward.

We introduce Nash Bargaining Preference Optimization (NBPO) for this setting. Within each objective, NBPO evaluates the learner against an adaptive opponent that emphasizes responses exposing the learner’s weaknesses, while a KL penalty keeps the opponent near the reference policy. Across objectives, it maximizes the Nash product of the resulting improvements over that same reference. The standard bargaining properties are useful in LLM alignment because they ensure that no objective falls below the fallback at the population optimum, no other feasible policy can improve one objective without worsening another, and arbitrary judge-specific units do not alter the selected compromise.

The main technical challenge is optimization, not re-deriving the classical bargaining axioms. Three issues arise simultaneously: the reference policy has zero improvement and lies on the boundary of the logarithmic objective; each adaptive opponent depends on the policy being optimized; and an LLM cannot directly realize the normalized finite-action policy update. We address them with a constrained KL-proximal formulation, a low-dimensional dual whose weights are inversely proportional to achieved improvements, a prompt-separable direct solve of the finite-pool inner problem, and partition-free pairwise log-ratio regression that projects the finite-pool target into the language model.

Our contributions are:

- We formulate multi-objective alignment as Nash bargaining over objective-wise regularized preference-game values, separating within-objective preference representation from across-objective compromise selection.
- We derive a practical optimizer that handles the zero-surplus boundary and policy-dependent opponents. For each dual iterate, the sampled finite-pool problem separates into small strictly concave prompt-wise programs; their target policy is transferred to the LLM through pairwise log-ratio regression without fitting a scalar reward or evaluating a language-model partition function.
- We prove $O(1/T)$ last-iterate convergence for exact population proximal updates. Feasibility-preserving controlled experiments isolate the benefit of adaptive game values under increasing nontransitivity, while a human-labeled SAFERLHF panel is reserved for testing Nash aggregation under natural objective conflict. Released natural datasets are not treated as evidence of within-objective cycles unless their comparison graphs identify such cycles.

2 BACKGROUND

General preference models. We use contextual-bandit notation. A prompt (context) x is drawn from a distribution $\mathcal{D}$, a response y is an action from a finite set $\mathcal{Y}_x$, and $\pi(y | x)$ is the policy. The reference policy is denoted by π_{ref} . For the finite-dimensional population analysis, the support of $\mathcal{D}$ and every $\mathcal{Y}_x$ are finite, and

$$\Pi = \prod_{x \in \text{supp}(\mathcal{D})} \Delta(\mathcal{Y}_x)$$

is the contextual policy simplex. Objective $k \in [K]$ is represented by a pairwise preference model $P_k(y \succ z \mid x) \in [0, 1]$ satisfying

$$P_k(y \succ z \mid x) + P_k(z \succ y \mid x) = 1, \quad P_k(y \succ y \mid x) = \frac{1}{2}. \quad (1)$$

We center each comparison at a tie,

$$\Delta_k(y, z \mid x) = P_k(y \succ z \mid x) - \frac{1}{2}, \quad \Delta_k(y, z \mid x) = -\Delta_k(z, y \mid x), \quad (2)$$

and define the expected objective- k payoff of learner policy π against comparator policy ν as

$$J_k(\pi, \nu) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi(\cdot \mid x), z \sim \nu(\cdot \mid x)} \Delta_k(y, z \mid x). \quad (3)$$

Thus $J_k(\pi, \nu) > 0$ means that π is preferred to ν on average under objective k . This representation does not assume a scalar reward or transitive preferences.

Bargaining problems. A K -dimensional bargaining problem consists of a feasible value set $\mathcal{G} \subseteq \mathbb{R}^K$ and a disagreement point $d \in \mathcal{G}$, the outcome obtained if no agreement is reached. For a feasible value vector v , define its objective-wise surplus over disagreement as

$$s_k(v) := v_k - d_k, \quad s(v) = (s_1(v), \dots, s_K(v)). \quad (4)$$

A value vector is individually rational when $s(v) \geq 0$ componentwise, and strictly individually rational when $s(v) > 0$. On a compact, convex, and downward-closed feasible set with at least one strictly individually rational point, the classical Nash solution is

$$v^N \in \arg \max_{v \in \mathcal{G}: s(v) > 0} \sum_{k=1}^K \log s_k(v). \quad (5)$$

Equivalently, it maximizes the product of the objective-wise surpluses (Nash, 1950).

3 OBJECTIVE-WISE PREFERENCE GAMES

We first convert each objective's pairwise preference model into a scalar policy value without imposing a scalar reward on individual responses. For objective k , an adaptive comparator ν searches for responses against which the learner performs poorly. The reference policy π_{ref} anchors this comparator and later serves as the fallback in the bargaining problem.

Definition 1 (Entropy-regularized game value). *For objective k and temperature $\beta_k > 0$, define*

$$V_{k, \beta_k}(\pi) = \min_{\nu} \{J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot \mid x) \parallel \pi_{\text{ref}}(\cdot \mid x))\}. \quad (6)$$

The comparator minimizes the learner's pairwise payoff, so it emphasizes responses that expose weaknesses of the current policy. The KL term prevents it from moving arbitrarily far from π_{ref} . Accordingly, a large β_k produces a local evaluation near the reference, whereas $\beta_k \downarrow 0$ approaches the unregularized worst-comparator value.

For a candidate comparator response z , let

$$m_k^\pi(z \mid x) := \mathbb{E}_{y \sim \pi(\cdot \mid x)} \Delta_k(y, z \mid x) \quad (7)$$

be the learner's expected margin against z . Smaller values identify stronger challenges. The inner problem in (6) has the closed-form solution

$$\nu_{k, \pi}^*(z \mid x) = \frac{\pi_{\text{ref}}(z \mid x) \exp\{-m_k^\pi(z \mid x)/\beta_k\}}{\mathbb{E}_{z' \sim \pi_{\text{ref}}(\cdot \mid x)} \exp\{-m_k^\pi(z' \mid x)/\beta_k\}}, \quad (8)$$

$$V_{k, \beta_k}(\pi) = -\beta_k \mathbb{E}_{x \sim \mathcal{D}} \log \mathbb{E}_{z \sim \nu_{k, \pi}^*(\cdot \mid x)} \exp\{-m_k^\pi(z \mid x)/\beta_k\}. \quad (9)$$

Thus the opponent is obtained by reweighting reference responses toward those with smaller margins, and the game value is a soft minimum of these margins.

Proposition 1 (Game-value structure). *For every k and $\beta_k \geq 0$, V_{k, β_k} is continuous and concave in π . For $\beta_k > 0$, its policy-gradient coordinate is*

$$[\nabla_{\pi} V_{k, \beta_k}(\pi)]_{x, y} = \mathcal{D}(x) Q_k^\pi(y \mid x), \quad Q_k^\pi(y \mid x) := \mathbb{E}_{z \sim \nu_{k, \pi}^*(\cdot \mid x)} \Delta_k(y, z \mid x). \quad (10)$$

The proof is in Appendix A.1. The response-level score $Q_k^\pi(y \mid x)$ is the expected margin of response y against the current objective- k opponent; it therefore measures the local effect of increasing $\pi(y \mid x)$. An adaptive opponent can distinguish policies that have the same win rate against one fixed reference but different vulnerabilities to other responses. Appendix A.2 gives a finite cyclic example at the unregularized endpoint $\beta = 0$; this illustration is not used in the later assumptions or proofs.

4 NASH BARGAINING OVER GAME-VALUE IMPROVEMENTS

We now combine the objective-wise game values using the reference policy as the common fallback. For each objective,

$$d_k := V_{k,\beta_k}(\pi_{\text{ref}}), \quad s_k(\pi) := V_{k,\beta_k}(\pi) - d_k. \quad (11)$$

The surplus $s_k(\pi)$ is the improvement of π over the deployed reference under objective k .

Assumption 1 (Joint improvement and support). *The reference π_{ref} has full support, and there exists a policy $\bar{\pi}$ such that $s_k(\bar{\pi}) > 0$ for every k .*

The assumption states that at least one policy improves all objectives over the fallback. Under this condition, the NBPO population target is

$$\pi^N \in \arg \max_{\pi \in \Pi_+} F(\pi), \quad F(\pi) := \sum_{k=1}^K \log s_k(\pi), \quad \Pi_+ := \{\pi \in \Pi : s_k(\pi) > 0, \forall k\}. \quad (12)$$

Equivalently, introducing auxiliary lower bounds $u_k > 0$ yields

$$(\pi^N, u^N) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \sum_{k=1}^K \log u_k \quad \text{s.t.} \quad u_k \leq s_k(\pi), \quad k \in [K], \quad \pi \in \Pi. \quad (13)$$

Because the objective is increasing in every u_k , each constraint is tight at an optimum and $u_k^N = s_k(\pi^N)$.

To connect (12) to the classical bargaining problem, define

$$V_{\beta}(\pi) = (V_{1,\beta_1}(\pi), \dots, V_{K,\beta_K}(\pi)), \quad d = (d_1, \dots, d_K),$$

and the downward-closed feasible value set

$$\mathcal{G}_{\beta} := \{v \in \mathbb{R}^K : d \leq v \leq V_{\beta}(\pi) \text{ componentwise for some } \pi \in \Pi\}. \quad (14)$$

Proposition 2 (Policy-induced Nash bargaining). *Under Assumption 1, $\mathcal{G}_{\beta}$ is compact, convex, downward closed above d , and nondegenerate. The policies solving (12) realize the classical Nash solution of $(\mathcal{G}_{\beta}, d)$. Consequently, every optimizer has positive surplus in every objective and is Pareto optimal in game-value space; all optimizers induce the same game-value vector; and the optimizer set is unchanged by independent transformations*

$$(V_{k,\beta_k}, d_k) \mapsto (c_k V_{k,\beta_k} + b_k, c_k d_k + b_k), \quad c_k > 0.$$

Appendix A.3 verifies the feasible-set conditions. Once those conditions hold, the stated outcome properties are standard consequences of the classical Nash bargaining solution (Nash, 1950); we do not claim them as new bargaining axioms. In LLM alignment, they mean that the population solution improves every objective over the reference, cannot be Pareto dominated by another feasible policy, and does not change merely because different judges report their values in different numerical units. Appendix A.4 compares these guarantees with utilitarian and maxmin aggregation.

5 NASH BARGAINING PREFERENCE OPTIMIZATION

Optimizing (12) presents three practical difficulties. First, the reference has $s_k(\pi_{\text{ref}}) = 0$, so the logarithmic objective is undefined at initialization. Second, every game value contains an opponent that depends on the learner. Third, the population policy update must be realized by an autoregressive model. We handle these issues with a constrained proximal problem, a prompt-separable finite-pool inner solve, and neural log-ratio projection.

5.1 CONSTRAINED PROXIMAL UPDATE AND ADAPTIVE OBJECTIVE WEIGHTS

For $\pi \in \Pi_+$, Proposition 1 and the chain rule give

$$[\nabla_{\pi} F(\pi)]_{x,y} = \mathcal{D}(x) \sum_{k=1}^K \frac{Q_k^{\pi}(y | x)}{s_k(\pi)}. \quad (15)$$

The inverse-surplus factors emphasize objectives that have improved less. Because this expression cannot be evaluated at π_{ref} , outer stage t instead solves

$$\begin{aligned} (\pi_{t+1}, u_{t+1}) \in \arg \max_{\pi, u \in \mathbb{R}_{++}^K} \quad & \sum_{k=1}^K \log u_k - \frac{1}{\eta_t} D(\pi \| \pi_t) \\ \text{s.t.} \quad & u_k \leq s_k(\pi), \quad k \in [K], \\ & \pi \in \Pi, \end{aligned} \quad (16)$$

where $\pi_0 = \pi_{\text{ref}}$, $\eta_t > 0$, and $D(\pi \| \pi_t) = \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\pi(\cdot | x) \| \pi_t(\cdot | x))$.

The dual is

$$\min_{\lambda \in \mathbb{R}_{++}^K} \phi_t(\lambda), \quad \phi_t(\lambda) = - \sum_{k=1}^K \log \lambda_k - K + \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}, \quad (17)$$

where $\lambda = (\lambda_1, \dots, \lambda_K) \in \mathbb{R}_{++}^K$ is the vector of Lagrange multipliers. For fixed λ , let

$$\pi_t^*(\lambda) := \arg \max_{\pi \in \Pi} \left\{ \sum_{k=1}^K \lambda_k s_k(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (18)$$

The KL term makes the maximizing policy $\pi_t^*(\lambda)$ unique. Danskin's theorem (Danskin, 1967) then allows us to differentiate the $\phi_t(\lambda)$ while holding this maximizing policy fixed, yielding

$$\nabla_k \phi_t(\lambda) = s_k(\pi_t^*(\lambda)) - \frac{1}{\lambda_k}. \quad (19)$$

The complete dual derivation is in Appendix A.5.

Proposition 3 (Inverse-surplus objective weights). *At an optimal primal–dual solution of (16),*

$$u_{t+1,k} = s_k(\pi_{t+1}), \quad \lambda_{t+1,k} = \frac{1}{u_{t+1,k}} = \frac{1}{s_k(\pi_{t+1})}. \quad (20)$$

Thus the constrained problem recovers the inverse-surplus weighting in (15) without selecting objective weights in advance.

5.2 DIRECT FINITE-POOL INNER SOLVE AND NEURAL PROJECTION

At one outer stage, sample learner candidates $\mathcal{Y}_x = \{y_{x,i}\}_{i=1}^{n_x}$ and comparator candidates $\mathcal{Z}_x = \{z_{x,j}\}_{j=1}^{m_x}$, and estimate the centered pairwise matrices $\hat{A}_{k,x}[i,j] = \hat{\Delta}_k(y_{x,i}, z_{x,j} | x)$. Let $p_{t,x}$ be the current learner mass on $\mathcal{Y}_x$ and let μ_x be the empirical reference mass on $\mathcal{Z}_x$. For $p \in \Delta^{n_x}$, define

$$\hat{m}_{k,x}(p)_j := \sum_{i=1}^{n_x} p_i \hat{A}_{k,x}[i,j], \quad (21)$$

$$\hat{V}_{k,x}(p) := -\beta_k \log \sum_{j=1}^{m_x} \mu_{x,j} \exp\{-\hat{m}_{k,x}(p)_j / \beta_k\}. \quad (22)$$

The disagreement value $\hat{d}_k$ is measured from an independently constructed reference–reference tensor rather than set to zero.

For fixed λ , the empirical counterpart of (18) separates across prompts:

$$p_{t,x}^*(\lambda) \in \arg \max_{p \in \Delta^{n_x}} \left\{ \sum_{k=1}^K \lambda_k \hat{V}_{k,x}(p) - \frac{1}{\eta_t} \text{KL}(p \| p_{t,x}) \right\}. \quad (23)$$

Each problem is strictly concave because of the KL term. We solve these small programs directly to a declared numerical tolerance and aggregate their values to obtain $\hat{s}_k(p_t^*(\lambda))$. The dual update is

$$\lambda^{(m+1)} = \Pi_\Lambda \left[\lambda^{(m)} - \gamma_m \left(\hat{s} \left(p_t^*(\lambda^{(m)}) \right) - (\lambda^{(m)})^{-1} \right) \right], \quad (24)$$

Algorithm 1 Nash Bargaining Preference Optimization

Require: Reference π_{ref} , pairwise models $P_{1:K}$, temperatures $\beta_{1:K}$, outer stages T .

- 1: Sample finite learner/comparator pools; construct pairwise and reference–reference tensors; estimate $\hat{d}_{1:K}$.
 - 2: Set $\pi_0 \leftarrow \pi_{\text{ref}}$.
 - 3: **for** $t = 0, \dots, T - 1$ **do**
 - 4: Score π_t on the learner pool and initialize positive λ .
 - 5: **repeat**
 - 6: Solve (23) independently for every prompt.
 - 7: Aggregate $\hat{s}_{1:K}$ and update λ by (24).
 - 8: **until** the projected dual/KKT criteria are met
 - 9: Construct $\hat{h}_t^*$ by (25) and verify (28).
 - 10: Fit one neural candidate $\tilde{\pi}_{t+1}$ using (27).
 - 11: Set $\pi_{t+1} \leftarrow \tilde{\pi}_{t+1}$ if held-out empirical surpluses are positive; otherwise retain π_t .
 - 12: **end for**
 - 13: **return** π_T .
-

where $\gamma_m > 0$ is the dual step size and Π_Λ denotes Euclidean projection onto $\Lambda = [\lambda_{\min}, \lambda_{\max}]^K$, with $0 < \lambda_{\min} < \lambda_{\max}$. This projection clips each multiplier to the prescribed interval, and the inverse is taken elementwise. In contrast, repeatedly alternating a closed-form opponent with the exponential policy map need not converge under strong nontransitivity; we retain that R -step map only as an ablation.

Once the dual converges, the finite-pool target defines the pairwise log-ratio change

$$\hat{h}_t^*(x, i, i') := \log \frac{p_{t,x}^*(i)}{p_{t,x}(i)} - \log \frac{p_{t,x}^*(i')}{p_{t,x}(i')}. \quad (25)$$

For a neural policy π_θ , let

$$h_t(\pi_\theta; x, y, y') := \log \frac{\pi_\theta(y | x)}{\pi_t(y | x)} - \log \frac{\pi_\theta(y' | x)}{\pi_t(y' | x)}. \quad (26)$$

We project the finite-pool target into the LLM with

$$\mathcal{L}_{\text{reg}}^{(t)}(\theta) = \mathbb{E}_{x, (i, i') \sim \xi_x} \left[h_t(\pi_\theta; x, y_{x,i}, y_{x,i'}) - \hat{h}_t^*(x, i, i') \right]^2. \quad (27)$$

No language-model partition function appears. At the inner optimum,

$$\hat{h}_t^*(x, i, i') = \eta_t \sum_{k=1}^K \lambda_k \left(\hat{Q}_{k,x}^*(i) - \hat{Q}_{k,x}^*(i') \right), \quad (28)$$

which provides an independent solver and artifact check. Pairwise probabilities may come from direct human comparisons or a general pairwise preference model; NBPO never requires them to decompose into scalar response rewards.

The finite-pool solve is numerical and the LLM projection remains approximate. The convergence result below concerns exact population solutions of (16).

5.3 LAST-ITERATE CONVERGENCE

Because Algorithm 1 returns the final outer policy rather than an average of past policies, we analyze the last population iterate.

Theorem 1 (Last-iterate convergence). *Under Assumption 1, let π^* maximize (12), set $\pi_0 = \pi_{\text{ref}}$, and solve (16) exactly at every outer iteration with arbitrary $\eta_t > 0$. Then $F(\pi_{t+1}) \geq F(\pi_t)$ for $t \geq 1$, and for every $T \geq 1$,*

$$F(\pi^*) - F(\pi_T) \leq \frac{D(\pi^* \| \pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (29)$$

A constant proximal step therefore gives an $O(1/T)$ final-iterate Nash-welfare gap. Appendix A.6 gives the proof.

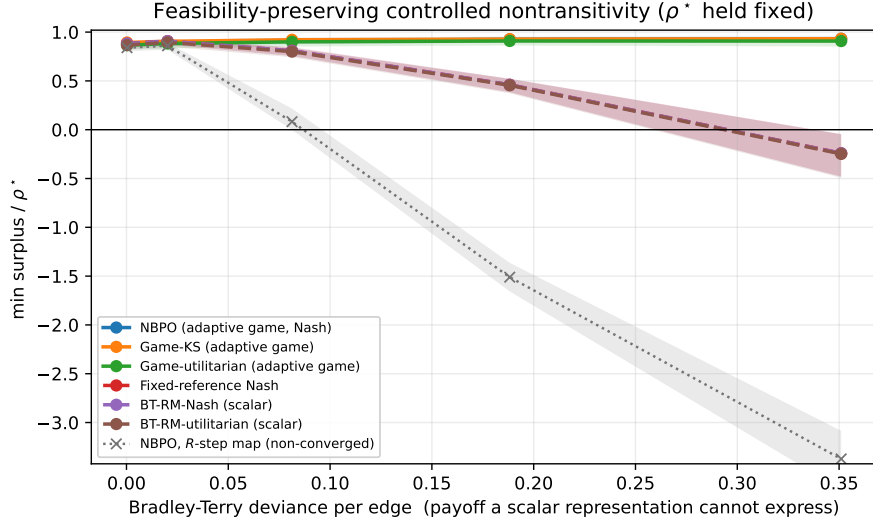


Figure 1: Feasibility-preserving controlled nontransitivity. Adaptive-game methods retain near-optimal worst-objective surplus as Bradley–Terry misspecification increases, whereas fixed-reference and scalar-reward representations deteriorate. The legacy R -step arm is shown only as a non-convergence ablation.

6 EXPERIMENTS

We separate the two empirical questions induced by the method. A feasibility-preserving controlled benchmark tests whether adaptive game values are needed as pairwise preferences become nontransitive. A human-labeled SAFERLHF panel tests whether Nash bargaining selects a useful compromise when helpfulness and harmlessness conflict. Finally, neural-realization and general-capability evaluations test whether the finite-pool target transfers to an 8B policy without unacceptable capability or safety regressions. This separation avoids using sparse released comparison graphs as evidence of natural within-objective cycles.

6.1 CONTROLLED NONTRANSITIVITY ISOLATES PREFERENCE REPRESENTATION

Construction. Each instance contains 40 prompts and four objectives. Starting from a transitive payoff component $B_{k,x}$, we add an odd-cardinality circulant skew-symmetric component $C_{k,x}$ whose zero row sums leave the reference disagreement value unchanged. We use α as the nontransitivity knob: schematically, $A_{k,x}^{(\alpha)} = B_{k,x} + \gamma(\alpha)C_{k,x}$, where the calibrated scale $\gamma(\alpha)$ increases the cyclic pair interaction as α increases. A bisection calibrates this scale so that the maximum attainable worst surplus

$$\rho^* := \max_{\pi} \min_k s_k(\pi)$$

remains approximately 0.03 while $\alpha \in \{0, .25, .5, .75, 1\}$ increases nontransitivity. No entrywise clipping is used. We run five independently generated seeds and evaluate every method with the same true adaptive-game values. The headline metric is $\min_k s_k(\pi)/\rho^*$, which normalizes the small remaining attainable-margin variation.

Representation, not an aggregation artifact. At $\alpha \leq .25$, all converged representations are similar. From $\alpha = .5$ onward, NBPO increasingly outperforms Fixed-reference Nash and BT-RM–Nash, reaching gaps of approximately 1.17 normalized units at $\alpha = 1$. BT-RM–Nash uses the same Nash rule as NBPO but collapses with its scalar-reward utilitarian counterpart, while NBPO, Game-KS, and game-utilitarian remain between 0.87 and 0.93. Thus the controlled result supports adaptive game-valued preference representation; it does not establish that Nash dominates every bargaining rule. Paired per-seed differences between NBPO and Game-KS do not reach significance at any α

Table 1: Controlled nontransitivity on the circulant feasibility-preserving family, averaged over five independent controlled seeds (not policy or preference-model seeds). Values are $\min_k s_k/\rho^*$; higher is better. The three adaptive-game rows remain stable, while the matched fixed-reference and scalar-reward controls fail as BT deviance increases. [†]The legacy R -step solver converged on 0/25 cells and is not a valid performance arm. Full objective-wise results and residuals are in Appendix B.

α	BT dev.	NBPO	Game-KS	Game-util.	Fixed-ref. Nash	BT-RM-Nash	BT-RM-util.	legacy R -step [†]
0.00	0.0003	0.881	0.893	0.866	0.883	0.883	0.867	0.840
0.25	0.0202	0.897	0.905	0.883	0.907	0.907	0.895	0.860
0.50	0.0813	0.914	0.921	0.900	0.809	0.809	0.800	0.082
0.75	0.1882	0.926	0.928	0.909	0.460	0.460	0.455	-1.511
1.00	0.3510	0.931	0.932	0.908	-0.241	-0.242	-0.248	-3.373

Table 2: Natural helpfulness-harmlessness bargaining on held-out SAFERLHF prompts. Objective columns are pairwise win rates against the common reference under the final independent evaluation protocol; cells report mean $\pm$ standard deviation over three policy-training seeds. “Conflict” evaluates the subset for which the released human objectives prefer different responses. Higher is better except HarmBench ASR.

method	helpful $\uparrow$	harmless $\uparrow$	worst $\uparrow$	average $\uparrow$	conflict worst $\uparrow$	HarmBench ASR $\downarrow$
Base = π_{ref}	0.500	0.500	0.500	0.500	—	—
NBPO (ours)	—	—	—	—	—	—
Fixed-reference Nash	—	—	—	—	—	—
BT-RM-Nash	—	—	—	—	—	—
Game-utilitarian	—	—	—	—	—	—
Game-KS	—	—	—	—	—	—
BT-RM-utilitarian	—	—	—	—	—	—
PROSPER/MaxEntBW [‡]	—	—	—	—	—	—

[‡]Retain only if a faithful matched-data run is completed; otherwise remove this row.

under a five-seed Student- t interval (for example -0.0007 , 95% CI $[-0.0347, +0.0332]$ at $\alpha = 1$); with five seeds this is an absence of evidence for a difference and not evidence of equivalence. The paired NBPO advantage over equal-weight game-utilitarian aggregation is positive in all 25 cells but reaches significance only at $\alpha = 0$ under the same interval.

Solver fidelity. On the controlled-v2 panel the direct prompt-wise solver meets every declared convergence gate in all 25 instances: projected KKT residual at most 3.6×10^{-15} , inner stationarity residual at most 6.6×10^{-9} against a 10^{-4} requirement, and finite-pool target identity at 1.8×10^{-15} . Table 6 decomposes exact, proximal and practical solutions on a *different* construction — the original interpolated family — where an exactly solved proximal reference is available; there the direct solve lies $\text{TV} = 0.027$ from that reference. Small inner residuals bound distance to the proximal solution at the current multipliers, not to a global Nash point. The legacy alternating map oscillates under stronger nontransitivity and fails every declared convergence gate. All final LLM experiments therefore require the direct finite-pool inner solve; Appendix E records the exact/proximal/practical decomposition and scaling measurements.

6.2 NATURAL OBJECTIVE CONFLICT ON SAFERLHF

SAFERLHF provides direct human pairwise labels for helpfulness and harmlessness (Dai et al., 2024). Its released graph contains no fully observed response triangles, so it cannot identify within-objective human cycles; we use it instead for the aggregation question. The two objectives disagree on 24.33% of held-out *human-annotated* pairs, which defines the conflict subset by a fixed prompt mapping. On *generated* response pools the frozen preference-model ensemble disagrees on only 7.7–7.8% of pairs (Table 10); the two rates have different populations, denominators and adjudicators and are not interchangeable. We fit matched-capacity objective models: an explicitly anti-symmetric joint-pair model for the general-preference interface and a scalar Bradley-Terry model for the reward-based controls. Their near-null held-out difference is reported rather than used as evidence for or against natural cyclicity. Final policy comparisons use the same sampled response pool and direct inner solver, three training seeds, and evaluation independent of the training preference models.

Table 3: General capability and safety transfer of the frozen SAFERLHF-trained checkpoints. Use official metrics where available; any local pairwise score must be named “reference win” rather than presented as the official benchmark metric. Mean $\pm$ standard deviation over three training seeds.

method	Arena-Hard ref. win	AlpacaEval-2 LC	IFEval strict	TruthfulQA MC2	HarmBench ASR $\downarrow$
Base = π_{ref}	0.500	—	—	—	—
NBPO (ours)	—	—	—	—	—
Fixed-reference Nash	—	—	—	—	—
BT-RM-Nash	—	—	—	—	—
Game-utilitarian	—	—	—	—	—
Game-KS	—	—	—	—	—

6.3 NEURAL REALIZATION AND GENERAL CAPABILITY

Before final seeds, a one-seed smoke test must show that the 8B policy learns the finite-pool log-ratio target: held-out normalized regression MSE below 0.90, sign agreement above 0.65, positive Pearson and Spearman correlations, and direct-solver residual below 10^{-4} . Final checkpoints are then evaluated on task-specific and general benchmarks without using the retired prompted-Qwen judge. In-domain SafeRLHF reward/cost models are labeled as diagnostics; deterministic metrics and blinded human comparisons provide complementary evidence.

7 RELATED WORK

General pairwise preferences. NLHF and its scalable variants optimize directly against general pairwise preference models rather than fitting scalar rewards (Munos et al., 2024; Wu et al., 2025; Zhang et al., 2025; Calandriello et al., 2024; Rosset et al., 2024; Zhou et al., 2025). Blackwell-style methods extend this view to vector-valued feedback and target sets (Blackwell, 1956; Bhatia et al., 2020). PROSPER introduces a maximum-entropy Blackwell winner with a scalable single-policy update and an inner worst-case choice over objective weights and opponents (Zhang et al., 2026). NBPO instead keeps one regularized game value per objective and applies Nash bargaining across the improvements over a shared fallback.

Reward-based multi-objective alignment. MODPO, Rewarded Soups, and Panacea expose user-specified trade-offs through scalarization, policy interpolation, or preference conditioning (Zhou et al., 2024; Ramé et al., 2023; Zhong et al., 2024b). MaxMin-RLHF and RMOD optimize weakest-objective criteria, while MOPO designates a primary objective and lower-bound constraints for the others (Chakraborty et al., 2024; Son et al., 2026; Agnihotri et al., 2026). These methods are important empirical baselines because they explicitly fit or use scalar objective rewards, whereas NBPO learns from general pairwise feedback without assuming that each objective is transitive.

Social choice and bargaining. The Nash solution and its Pareto, uniqueness, and affine-invariance properties are classical (Nash, 1950). Social-choice work argues that alignment should state its aggregation rule explicitly (Conitzer et al., 2024). Multi-party RLHF studies utilitarian, Nash, and leximin welfare over party-specific learned rewards and separately considers reward-free von Neumann solutions (Zhong et al., 2024a); Nash bargaining has also been used to combine task gradients in multi-task learning (Navon et al., 2022). Our contribution is not a new bargaining axiom, but a formulation and practical optimizer for applying the classical Nash rule to policy-dependent objective-wise preference-game values.

8 CONCLUSION AND LIMITATIONS

NBPO separates two decisions in multi-objective alignment: adaptive game values represent general pairwise preferences within each objective, and Nash bargaining selects one reference-relative compromise across objectives. The constrained proximal formulation inherits the classical Nash outcome properties and admits an $O(1/T)$ exact-population last-iterate guarantee. Practically, a direct prompt-separable finite-pool inner solve is necessary: it reproduces the exact proximal target in the controlled study, whereas a naive alternating opponent-policy map can oscillate under strong nontransitivity. Feasibility-preserving experiments support the representation claim by showing that

adaptive-game methods remain stable as scalar-reward and fixed-reference representations deteriorate. The natural SAFERLHF panel separately tests whether Nash improves the helpfulness–harmlessness compromise.

The evidence has clear limits. The controlled nontransitivity study is constructed rather than a measurement of natural human cycles. Existing released datasets used here do not densely compare response triples, so they cannot establish within-objective human intransitivity. The SafeRLHF preference models may face distribution shift on newly generated responses, and the neural policy only approximates the finite-pool target. Nash is not empirically superior to Game-KS in the controlled panel, and any natural aggregation claim must be supported by the completed multi-seed and independent-evaluation results in Table 2. Finally, Assumption 1 can fail when no policy improves every objective over the reference; game-value improvements are average quantities, not per-prompt safety guarantees.

AI USE STATEMENT

In this work, generative AI tools were used to critique the problem formulation, assist with mathematical derivations and proof presentation, propose experimental checks, and edit manuscript text. The authors are responsible for independently verifying every theorem, citation, implementation, empirical result, and disclosure in the final submission.

ETHICS STATEMENT

This work studies aggregation rules for language-model alignment, including harmlessness as one objective. Improvement in a game value is not an absolute safety certificate. Safety-critical deployment should retain hard constraints and independent safety evaluations in addition to the bargaining objective.

REPRODUCIBILITY STATEMENT

Algorithm 1 distinguishes the population proximal problem, the numerically solved finite-pool inner problem, and its neural projection. Appendix A contains the dual and convergence proofs; Appendices B–G specify the controlled construction, natural supervision, hyperparameters, solver and neural diagnostics, and the prompted-judge audit. The submission package should include candidate pools, preference tensors, dual traces, solver residuals, per-seed outputs, and the code and configurations required to reproduce the reported results.

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A PROOFS

A.1 PROOF OF PROPOSITION 1

For a fixed comparator ν , define

$$\mathcal{J}_{k,\nu}(\pi) := J_k(\pi, \nu) + \beta_k \mathbb{E}_{x \sim \mathcal{D}} \text{KL}(\nu(\cdot | x) \| \pi_{\text{ref}}(\cdot | x)).$$

The first term is linear in π and the second does not depend on π , so $\mathcal{J}_{k,\nu}$ is affine. The pointwise minimum of affine functions is concave; hence V_{k,β_k} is concave. Since $|\Delta_k| \leq 1/2$,

$$|V_{k,\beta_k}(\pi) - V_{k,\beta_k}(\pi')| \leq \frac{1}{2} \mathbb{E}_{x \sim \mathcal{D}} \|\pi(\cdot | x) - \pi'(\cdot | x)\|_1,$$

which also proves continuity.

For $\beta_k > 0$, the inner minimization separates across prompts. Fix x and abbreviate $m(z) = m_k^\pi(z | x)$. The prompt-wise problem is

$$\min_{\nu \in \Delta(\mathcal{Y}_x)} \sum_z \nu(z) m(z) + \beta_k \sum_z \nu(z) \log \frac{\nu(z)}{\pi_{\text{ref}}(z | x)}.$$

The first-order condition under $\sum_z \nu(z) = 1$ gives

$$\nu(z) \propto \pi_{\text{ref}}(z | x) \exp\{-m(z)/\beta_k\},$$

which yields (8); substituting it back yields (9). The KL term makes the minimizer unique. Danskin's theorem therefore gives, for every feasible direction h satisfying $\sum_y h(y | x) = 0$,

$$DV_{k,\beta_k}(\pi)[h] = \mathbb{E}_{x \sim \mathcal{D}} \sum_y h(y | x) \mathbb{E}_{z \sim \nu_{k,\pi}^*(\cdot | x)} \Delta_k(y, z | x).$$

Thus the coordinate derivative is

$$[\nabla_\pi V_{k,\beta_k}(\pi)]_{x,y} = \mathcal{D}(x) Q_k^\pi(y | x),$$

proving (10).

A.2 CONNECTION TO THE FIXED-REFERENCE

This subsection gives a finite-action illustration at the unregularized endpoint $\beta = 0$. It is not used in Assumption 1, the full-support analysis, or the finite-temperature experiments.

Define the fixed-reference improvement

$$a_k(\pi) = J_k(\pi, \pi_{\text{ref}}) = P_k(\pi \succ \pi_{\text{ref}}) - \frac{1}{2}. \quad (30)$$

Skew-symmetry gives $a_k(\pi_{\text{ref}}) = 0$.

Full cyclic example. For one prompt, let $\pi_{\text{ref}} = \delta_r$, where δ_y is the deterministic policy that always outputs y . Suppose A , B , and C each beat r with probability 0.75 and form the deterministic cycle

$$A \succ B, \quad B \succ C, \quad C \succ A.$$

Both

$$\pi_{\text{pure}} = \delta_A \quad \text{and} \quad \pi_{\text{mix}} = \frac{\delta_A + \delta_B + \delta_C}{3}$$

have the same fixed-reference improvement,

$$a(\pi_{\text{pure}}) = a(\pi_{\text{mix}}) = 0.75 - \frac{1}{2} = 0.25.$$

Their worst-opponent values are nevertheless different. For the pure policy, response C defeats A deterministically, so

$$V_0(\pi_{\text{pure}}) = \min_{\nu} J(\pi_{\text{pure}}, \nu) = J(\delta_A, \delta_C) = -\frac{1}{2}.$$

For the uniform mixture, consider first the pure opponent δ_A . The mixture outputs A , B , and C with equal probability. Against A , these responses respectively tie, lose, and win, giving centered payoffs 0, $-1/2$, and $+1/2$. Hence

$$J(\pi_{\text{mix}}, \delta_A) = \frac{1}{3} \left(0 - \frac{1}{2} + \frac{1}{2} \right) = 0.$$

By symmetry,

$$J(\pi_{\text{mix}}, \delta_B) = J(\pi_{\text{mix}}, \delta_C) = 0.$$

Against the reference,

$$J(\pi_{\text{mix}}, \delta_r) = 0.25,$$

because every response in the mixture beats r with probability 0.75. Since $J(\pi_{\text{mix}}, \nu)$ is linear in ν , no mixed opponent can achieve a value below the minimum over these pure opponents. Therefore

$$V_0(\pi_{\text{mix}}) = \min_{\nu} J(\pi_{\text{mix}}, \nu) = \min\{0, 0, 0, 0.25\} = 0.$$

Finally, the reference itself has

$$V_0(\pi_{\text{ref}}) = \min_{\nu} J(\delta_r, \nu) = -0.25,$$

since r loses to each of A , B , and C with probability 0.75. Thus

	$a(\pi)$	$V_0(\pi)$	$V_0(\pi) - V_0(\pi_{\text{ref}})$
π_{pure}	0.25	-0.50	-0.25
π_{mix}	0.25	0	0.25

The fixed reference therefore assigns the two policies the same score, even though the pure policy is fully exploitable by one response whereas the balanced mixture guarantees an average tie against every response in the cycle.

A.3 POLICY-INDUCED NASH BARGAINING AND CLASSICAL CONSEQUENCES

We prove Proposition 2. Since Π is compact and each V_{k,β_k} is continuous, $V_\beta(\pi)$ is bounded. The definition of $\mathcal{G}_\beta$ then implies boundedness. Closedness follows by taking any convergent sequence $v_n \in \mathcal{G}_\beta$, choosing policies π_n with $d \leq v_n \leq V_\beta(\pi_n)$, extracting a convergent policy subsequence, and applying continuity. Downward closure is immediate from the definition.

For convexity, take $v^1, v^2 \in \mathcal{G}_\beta$ with witnesses π_1, π_2 . Objective-wise concavity from Proposition 1 gives, for $\alpha \in [0, 1]$,

$$V_\beta(\alpha\pi_1 + (1 - \alpha)\pi_2) \geq \alpha V_\beta(\pi_1) + (1 - \alpha)V_\beta(\pi_2)$$

componentwise, so $\alpha v^1 + (1 - \alpha)v^2 \in \mathcal{G}_\beta$. Assumption 1 provides a point strictly above d , proving nondegeneracy.

For any $v \in \mathcal{G}_\beta$ with $v > d$, some policy π satisfies $v \leq V_\beta(\pi)$. Since the Nash objective is coordinatewise increasing,

$$\sum_k \log(v_k - d_k) \leq \sum_k \log(V_{k,\beta_k}(\pi) - d_k) = F(\pi).$$

Conversely, every $\pi \in \Pi_+$ induces a feasible value vector $V_\beta(\pi)$. Hence (12) realizes the classical Nash solution over $(\mathcal{G}_\beta, d)$.

The remaining statements are standard Nash-bargaining consequences, but we record the short arguments for completeness. Since the log objective tends to $-\infty$ when any surplus approaches zero, every optimizer lies strictly above d . Coordinatewise monotonicity implies Pareto optimality. Strict concavity of $\sum_k \log s_k$ over value vectors implies that all optimal policies induce the same surplus, hence the same game-value vector, although the policy itself can be nonunique.

Finally, under independent positive affine transformations

$$V'_k(\pi) = c_k V_k(\pi) + b_k, \quad d'_k = c_k d_k + b_k, \quad c_k > 0,$$

the transformed surplus is $s'_k(\pi) = c_k s_k(\pi)$ and

$$\sum_k \log s'_k(\pi) = \sum_k \log s_k(\pi) + \sum_k \log c_k.$$

The added term is independent of π , so the optimizer set is unchanged.

Table 4: Classical population guarantees of common aggregation rules under joint improvement. The Nash row follows from the standard bargaining solution; the other rows are supported by the finite counterexamples below.

rule	strict IR	Pareto optimal	unique value	affine invariant
positive-weight utilitarian	no	yes	no	no
absolute maxmin	no	no	no	no
surplus maxmin	yes	no	no	no
Nash bargaining	yes	yes	yes	yes

A.4 COUNTEREXAMPLES FOR ALTERNATIVE AGGREGATION RULES

The positive guarantees in Table 4 follow directly from the corresponding objectives. Because $w_k > 0$, the positive-weight utilitarian objective $\sum_k w_k V_k(\pi)$ is strictly increasing in every game-value coordinate, so none of its optimizers can be Pareto dominated. Under Assumption 1, surplus maxmin also attains a strictly positive minimum surplus because a jointly improving policy exists. The examples below establish the guarantees that can fail for the alternative rules.

All examples use three responses, so a policy is a probability vector $\pi = (\pi_1, \pi_2, \pi_3) \in \Delta_3$, and use the uniform reference $\pi_{\text{ref}} = (1/3, 1/3, 1/3)$. For objective k , Δ_k is the skew-symmetric pairwise payoff matrix: $(\Delta_k)_{ij}$ is the centered preference margin of response i against response j . At $\beta_k = 0$,

$$V_k(\pi) = \min_j (\pi^\top \Delta_k)_j, \quad d_k = V_k(\pi_{\text{ref}}), \quad s_k(\pi) = V_k(\pi) - d_k.$$

Thus $\pi^\top \Delta_k$ contains the expected payoff of π against each pure opponent, and $V_k(\pi)$ is its smallest coordinate.

Because each V_k is the minimum of linear functions, the optimizers below are obtained from small linear programs. For example, equal-weight utilitarian optimization can be written as

$$\max_{\pi, t_1, t_2} t_1 + t_2 \quad \text{s.t.} \quad t_k \leq (\pi^\top \Delta_k)_j \quad \forall k, j, \quad \pi \in \Delta_3.$$

Absolute maximin uses one variable t with $t \leq (\pi^\top \Delta_k)_j$, while surplus maximin uses $t + d_k \leq (\pi^\top \Delta_k)_j$ for every k, j .

Utilitarian failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & -1/2 \\ 1/2 & 0 & -1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & -1/2 \\ -1/4 & 1/2 & 0 \end{pmatrix}.$$

For the uniform reference,

$$\pi_{\text{ref}}^\top \Delta_1 = (1/3, 0, -1/3), \quad \pi_{\text{ref}}^\top \Delta_2 = (1/12, 0, -1/12),$$

and hence

$$d = (-1/3, -1/12).$$

First, joint improvement is feasible. The policy

$$\bar{\pi} = (1/4, 1/4, 1/2)$$

is only a feasibility witness; it gives

$$\bar{\pi}^\top \Delta_1 = (3/8, 1/8, -1/4), \quad \bar{\pi}^\top \Delta_2 = (0, 1/8, -1/16).$$

Therefore

$$V(\bar{\pi}) = (-1/4, -1/16), \quad s(\bar{\pi}) = V(\bar{\pi}) - d = (1/12, 1/48) > 0.$$

In contrast, solving the equal-weight utilitarian problem $\max_{\pi} V_1(\pi) + V_2(\pi)$ gives

$$\pi_U = (0, 1/5, 4/5).$$

For this policy,

$$\pi_U^\top \Delta_1 = (1/2, 2/5, -1/10), \quad \pi_U^\top \Delta_2 = (-1/10, 2/5, -1/10),$$

so

$$V(\pi_U) = (-1/10, -1/10), \quad s(\pi_U) = (7/30, -1/60).$$

Thus joint improvement is possible, yet a utilitarian optimum falls below the fallback on objective 2.

Absolute-maximin failure of strict improvement. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/4 \\ 1/2 & 0 & 1/2 \\ -1/4 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 1/4 & -1/2 \\ -1/4 & 0 & 1/2 \\ 1/2 & -1/2 & 0 \end{pmatrix}.$$

Again,

$$d = (-1/3, -1/12).$$

Joint improvement is feasible, as witnessed by

$$\bar{\pi} = (1/3, 7/15, 1/5),$$

for which

$$V(\bar{\pi}) = (-4/15, -1/60)$$

and therefore

$$s(\bar{\pi}) = (1/15, 1/15) > 0.$$

Absolute maxmin instead solves

$$\max_{\pi} \min\{V_1(\pi), V_2(\pi)\},$$

whose linear-program solution is

$$\pi_{\text{AM}} = (0, 4/5, 1/5).$$

For this policy,

$$\pi_{\text{AM}}^{\top} \Delta_1 = (7/20, -1/10, 2/5), \quad \pi_{\text{AM}}^{\top} \Delta_2 = (-1/10, -1/10, 2/5),$$

so

$$V(\pi_{\text{AM}}) = (-1/10, -1/10)$$

and

$$s(\pi_{\text{AM}}) = (7/30, -1/60).$$

Thus absolute maxmin maximizes the smallest *absolute* game value while allowing objective 2 to fall below its own fallback.

A Pareto-dominated surplus-maxmin optimum. Let

$$\Delta_1 = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}, \quad \Delta_2 = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 1/2 \\ -1/2 & -1/2 & 0 \end{pmatrix}.$$

The uniform reference gives

$$d = (-1/3, -1/6).$$

For objective 2, no policy can achieve $V_2(\pi) > 0$, while every mixture supported on the first two responses achieves $V_2(\pi) = 0$. Consequently,

$$s_2(\pi) \leq 1/6$$

for every policy, so the optimal surplus-maxmin value cannot exceed $1/6$.

The two policies

$$\pi_c = (1/3, 2/3, 0), \quad \pi_e = (0, 1, 0)$$

satisfy

$$V(\pi_c) = (-1/6, 0), \quad s(\pi_c) = (1/6, 1/6),$$

and

$$V(\pi_e) = (0, 0), \quad s(\pi_e) = (1/3, 1/6).$$

Both therefore maximize

$$\min_k s_k(\pi)$$

at $1/6$. However,

$$s(\pi_e) > s(\pi_c)$$

in the first coordinate and is equal in the second, so π_e Pareto dominates π_c . Thus the surplus-maxmin optimizer set can contain Pareto-dominated policies.

A.5 DUAL DERIVATION AND REGRESSION FIXED POINT

For multipliers $\lambda_k \geq 0$, the Lagrangian of (16) is

$$\mathcal{L}_t(\pi, u, \lambda) = \sum_k \log u_k + \sum_k \lambda_k (s_k(\pi) - u_k) - \eta_t^{-1} D(\pi \| \pi_t).$$

For fixed $\lambda_k > 0$, maximizing $\log u_k - \lambda_k u_k$ gives $u_k = 1/\lambda_k$ and value $-\log \lambda_k - 1$; if $\lambda_k = 0$, the maximum is unbounded. Substitution yields (17), and the remaining maximization over π is (18). Uniqueness of that maximizer and Danskin's theorem yield (19).

Assumption 1 provides strict feasibility: choose $\bar{\pi}$ with $s_k(\bar{\pi}) > 0$ and $0 < \bar{u}_k < s_k(\bar{\pi})$. Slater's condition therefore gives strong duality. Stationarity in u_k gives

$$\lambda_{t+1,k} = \frac{1}{u_{t+1,k}} > 0,$$

and complementary slackness then forces $u_{t+1,k} = s_k(\pi_{t+1})$. This proves Proposition 3.

Finite-pool optimality and neural projection. Fix λ and let $\pi_t^*(\lambda)$ solve (18). First-order optimality gives, for every prompt and response pair in the support,

$$h_t(\pi_t^*(\lambda); x, y, y') = \eta_t \sum_k \lambda_k \left(Q_k^{\pi_t^*(\lambda)}(y | x) - Q_k^{\pi_t^*(\lambda)}(y' | x) \right). \quad (31)$$

The direct finite-pool program (23) is the sampled analogue of this strictly concave problem. Its normalized optimizer determines all finite-pool log-probability differences, so (25) is identified without computing a language-model normalizer. Equation (28) is the corresponding finite-pool KKT identity and is checked numerically before an artifact is accepted. With a full-support pair proposal, a zero-regression-error minimizer of (27) matches every target log-probability difference; normalization therefore recovers the finite-pool target distribution on the sampled support. This statement concerns the regression target, not the finite-sample optimization error of the neural model.

A.6 PROOF OF THEOREM 1

Let $h(\pi) = \mathbb{E}_{x \sim \mathcal{D}} \sum_y \pi(y | x) \log \pi(y | x)$, so its Bregman divergence is $D(\pi \| \pi_t)$. Because the objective is increasing in u , the policy part of (16) is equivalently

$$\pi_{t+1} \in \arg \max_{\pi \in \Pi_+} \left\{ F(\pi) - \frac{1}{\eta_t} D(\pi \| \pi_t) \right\}. \quad (32)$$

The feasible set is nonempty by Assumption 1, and $F(\pi) \rightarrow -\infty$ as any surplus approaches zero. Thus the maximizer exists in Π_+ even when the center π_t is the reference $\pi_{\text{ref}} \notin \Pi_+$. Because π_t has full support and the remaining objective has finite directional derivatives at an interior surplus point, the inward derivative of $-D(\pi \| \pi_t)$ excludes zero-probability coordinates. Induction therefore gives a full-support, positive-surplus π_{t+1} .

Let g_{t+1} be a supergradient of F at π_{t+1} . First-order optimality of (32) and concavity of F imply, for every $\pi \in \Pi_+$,

$$F(\pi) - F(\pi_{t+1}) \leq \langle g_{t+1}, \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (33)$$

$$\leq \frac{1}{\eta_t} \langle \nabla h(\pi_{t+1}) - \nabla h(\pi_t), \pi - \pi_{t+1} \rangle_{\mathcal{D}} \quad (34)$$

$$= \frac{1}{\eta_t} [D(\pi \| \pi_t) - D(\pi \| \pi_{t+1}) - D(\pi_{t+1} \| \pi_t)]. \quad (35)$$

For $t \geq 1$, π_t is feasible in the next subproblem. Comparing the optimum with π_t gives

$$F(\pi_{t+1}) - \frac{1}{\eta_t} D(\pi_{t+1} \| \pi_t) \geq F(\pi_t), \quad (36)$$

so the objective values are nondecreasing from π_1 onward.

Set $\pi = \pi^*$ in (35), multiply by η_t , and sum from $t = 0$ to $T - 1$:

$$\sum_{t=0}^{T-1} \eta_t (F(\pi^*) - F(\pi_{t+1})) \leq D(\pi^* \| \pi_{\text{ref}}). \quad (37)$$

Because the gaps are nonincreasing, the left-hand side is at least $(\sum_t \eta_t)(F(\pi^*) - F(\pi_T))$, which proves (29).

Corollary 1 (Convergence of the surplus profile). *Under the conditions of Theorem 1, let $s^* = s(\pi^*)$ denote the unique surplus vector induced by a Nash-optimal policy. Then*

$$\|s(\pi_T) - s^*\|_2^2 \leq \frac{2D(\pi^* \| \pi_{\text{ref}})}{\sum_{t=0}^{T-1} \eta_t}. \quad (38)$$

Hence a constant proximal step gives $\|s(\pi_T) - s^*\|_2 = O(T^{-1/2})$.

Proof. This is a stronger consequence of Nash-welfare convergence rather than a separate requirement for the NBPO update. For every objective, $V_k(\pi) \in [-1/2, 1/2]$, and $d_k = V_k(\pi_{\text{ref}}) \leq 0$ because

Table 5: Full controlled-v2 results, averaged over five seeds. “Solver resid.” is the method-appropriate inner or KKT residual; an entry of zero for a non-adaptive method does not certify an adaptive fixed point. The legacy R -step arm is a non-convergence ablation.

method	min surplus	sd	min s/ρ^*	sd	exploitability	inner resid.
$\alpha = 0.00, \rho^* = 0.0300, BT\ dev. = 0.0003$						
NBPO direct	+0.0264	0.0011	0.881	0.037	0.0687	2×10^{-12}
Game-KS	+0.0268	0.0005	0.893	0.017	0.0685	10^{-9}
Game-util.	+0.0260	0.0014	0.866	0.048	0.0688	3×10^{-10}
Fixed-ref. Nash	+0.0265	0.0011	0.883	0.036	0.0688	0
BT-RM-Nash	+0.0265	0.0011	0.883	0.036	0.0688	0
BT-RM-util.	+0.0260	0.0014	0.867	0.047	0.0689	0
legacy R -step [†]	+0.0252	0.0012	0.840	0.041	0.0675	5×10^{-10}
$\alpha = 0.25, \rho^* = 0.0300, BT\ dev. = 0.0202$						
NBPO direct	+0.0269	0.0013	0.897	0.044	0.1091	10^{-7}
Game-KS	+0.0272	0.0007	0.905	0.025	0.1088	10^{-7}
Game-util.	+0.0265	0.0017	0.883	0.058	0.1094	4×10^{-8}
Fixed-ref. Nash	+0.0272	0.0012	0.907	0.040	0.1197	0
BT-RM-Nash	+0.0272	0.0012	0.907	0.040	0.1197	0
BT-RM-util.	+0.0269	0.0015	0.895	0.049	0.1198	0
legacy R -step [†]	+0.0258	0.0014	0.860	0.046	0.1063	7×10^{-1}
$\alpha = 0.50, \rho^* = 0.0300, BT\ dev. = 0.0813$						
NBPO direct	+0.0274	0.0013	0.914	0.044	0.1454	2×10^{-6}
Game-KS	+0.0276	0.0007	0.921	0.023	0.1450	2×10^{-7}
Game-util.	+0.0270	0.0017	0.900	0.057	0.1458	2×10^{-7}
Fixed-ref. Nash	+0.0243	0.0018	0.809	0.059	0.1945	0
BT-RM-Nash	+0.0243	0.0018	0.809	0.058	0.1945	0
BT-RM-util.	+0.0240	0.0019	0.800	0.064	0.1947	0
legacy R -step [†]	+0.0025	0.0044	0.082	0.147	0.1981	10×10^{-1}
$\alpha = 0.75, \rho^* = 0.0300, BT\ dev. = 0.1882$						
NBPO direct	+0.0278	0.0013	0.926	0.042	0.1693	8×10^{-6}
Game-KS	+0.0278	0.0005	0.928	0.018	0.1692	8×10^{-7}
Game-util.	+0.0273	0.0017	0.909	0.056	0.1696	7×10^{-7}
Fixed-ref. Nash	+0.0138	0.0027	0.460	0.089	0.2723	0
BT-RM-Nash	+0.0138	0.0027	0.460	0.089	0.2723	0
BT-RM-util.	+0.0137	0.0028	0.455	0.095	0.2724	0
legacy R -step [†]	-0.0453	0.0060	-1.511	0.201	0.2626	10×10^{-1}
$\alpha = 1.00, \rho^* = 0.0283, BT\ dev. = 0.3510$						
NBPO direct	+0.0265	0.0033	0.931	0.041	0.1824	5×10^{-5}
Game-KS	+0.0264	0.0027	0.932	0.019	0.1825	3×10^{-6}
Game-util.	+0.0258	0.0037	0.908	0.063	0.1826	2×10^{-6}
Fixed-ref. Nash	-0.0063	0.0068	-0.241	0.277	0.3464	0
BT-RM-Nash	-0.0063	0.0068	-0.242	0.276	0.3464	0
BT-RM-util.	-0.0065	0.0070	-0.248	0.284	0.3465	0
legacy R -step [†]	-0.0949	0.0064	-3.373	0.406	0.3337	10^0

$\nu = \pi_{\text{ref}}$ is feasible and $J_k(\pi_{\text{ref}}, \pi_{\text{ref}}) = 0$. Hence every positive surplus is at most one. On $(0, 1]^K$, the function $G(s) = \sum_k \log s_k$ is 1-strongly concave in Euclidean norm. Proposition 2 implies that all Nash-optimal policies induce the same game-value vector, and therefore the same surplus vector s^* . First-order optimality over the convex comprehensive surplus set induced by (14) gives

$$G(s^*) - G(s) \geq \frac{1}{2} \|s - s^*\|_2^2. \quad (39)$$

Applying this inequality to $s = s(\pi_T)$ and using (29) proves (38). $\square$

B CONTROLLED BENCHMARK: CONSTRUCTION AND FULL RESULTS

Feasibility-preserving construction. For every seed we form a transitive base matrix and an odd-cardinality circulant skew-symmetric perturbation. Zero row sums keep the uniform reference disagreement invariant, while bisection selects the cycle magnitude so that $\rho^* = \max_{\pi} \min_k s_k(\pi)$ stays close to a target margin. We reject invalid probabilities rather than clip matrix entries. The resulting panel increases BT deviance from approximately 2.7×10^{-4} to 0.351 while retaining positive joint improvement in every instance. All policies are evaluated under the same adaptive-game values, not their method-native scales.

Table 6: Finite-pool solver audit at $\alpha = 1$ in the original controlled family. The direct practical solve recovers the positive-surplus proximal target; the deployed legacy map does not.

solution	min surplus	$/\rho^*$	inner residual	TV to exact prox.
Exact global Nash	+0.0158	+0.935	–	0.007
Exact proximal Nash	+0.0158	+0.936	–	0.000
NBPO direct inner solve	+0.0139	+0.821	2×10^{-6}	0.027
Legacy R -step map	–0.1851	–10.938	10^0	0.643
BT-RM–Nash	–0.1001	–5.918	0	0.495

Table 7: Robustness of the controlled trend. Gaps are the paired per-seed difference in $\min_k s_k/\rho^*$ at $\alpha = 1$, with an exact Student- t 95% interval over seeds (*not* a bootstrap interval). The circulant row and the projected skew-symmetric rows are different constructions; the β sweep was run on the projected family only. Converged counts exclude the legacy R -step ablation, which converges on no cell in any run.

cycle family	β	seeds	NBPO–FixedRef (t_{95})	NBPO–BT-RM (t_{95})	min ρ^*	converged
Circulant (primary)	0.25	5	+1.1724 [+0.8715, +1.4733]	+1.1733 [+0.8735, +1.4731]	0.0283	150/150
Projected skew-symmetric	0.25	5	+0.1114 [+0.0714, +0.1514]	+0.1115 [+0.0715, +0.1514]	0.0332	150/150
Projected skew-symmetric	0.10	3	+0.2381 [+0.0880, +0.3883]	+0.2385 [+0.0883, +0.3887]	0.0331	90/90
Projected skew-symmetric	0.50	3	+0.0448 [+0.0165, +0.0730]	+0.0449 [+0.0167, +0.0730]	0.0334	90/90

Robustness across construction and temperature. The primary construction is checked against an independently generated projected skew-symmetric family, $C = N(R - R^\top)N$ with N the symmetric projector annihilating both μ and a witness policy, and against a temperature sweep on that family. The trend replicates: adaptive-game rows stay flat while fixed-reference and scalar-reward rows fall. The effect *size* does not: the $\alpha = 1$ gap is +0.111 on the projected family against +1.172 on the circulant one, because the projected family’s BT deviance only reaches 0.174 against 0.351. We therefore claim robustness of the trend and do not attach the circulant effect size to it. Intervals in Table 7 are exact paired Student- t intervals over seeds, not bootstrap intervals.

C NATURAL SUPERVISION AND PREFERENCE MODELS

Dataset audit. Table 8 records the empirical role of each released dataset. ULTRAFeEDBACK supplies complete per-completion ordinal attributes, so score-induced pairwise labels are transitive by construction. SAFERLHF supplies direct helpfulness and safety comparisons but almost exclusively isolated edges; zero observed triangles means the data cannot test within-objective cycles. It nevertheless contains substantial cross-objective disagreement and is therefore appropriate for the bargaining experiment.

Anti-symmetric general preference model. For objective k , a joint encoder produces $a_{\phi,k}(x, y, z)$ and we set

$$\ell_{\phi,k}(x, y, z) = \frac{1}{2}\{a_{\phi,k}(x, y, z) - a_{\phi,k}(x, z, y)\}, \quad P_{\phi,k}(y \succ z \mid x) = \sigma(\ell_{\phi,k}(x, y, z)).$$

This guarantees exact swap complementarity and self-ties without constraining the logit to a scalar reward difference. In the SafeRLHF diagnostic, both the joint model and the scalar BT baseline use RoBERTa-base so that encoder capacity and training cost are matched; this backbone choice is an experimental control rather than part of the NBPO method. Because the observed SafeRLHF graph contains no triangles, their near-null comparison is a calibration result rather than evidence that human preferences are transitive.

D FINAL TRAINING PROTOCOL AND HYPERPARAMETERS

All method comparisons share prompts, candidate pools, preference tensors, tokenizer and sequence masks, policy-training budget, decoding, and evaluation. The core rows vary only the objective representation or aggregation rule: adaptive game versus fixed-reference margin versus scalar BT

Table 8: Released-supervision audit. “Obs. triangles” counts fully observed closed triples of *human* comparisons, summed over both objectives and all three splits; zero means within-objective human cycles are unmeasurable here, which is not a measured cycle rate of zero. “Human conflict” is the released-label disagreement rate on the held-out split.

dataset	rows	prompts	responses	obs. triangles	human conflict	supervision	role
SAFERLHF	73,907	38,639	143,668	0	24.33%	direct pairwise helpful/safe	natural objective conflict
ULTRAFEEDBACK	63,967	63,967	4/prompt	n/a (score-induced)	n/a	four ordinal attributes	transitive aggregation control

Table 9: Held-out preference-model validation, seeds 41/42/43. All values come from the *checkpointed* training run whose weights are on disk and which every downstream scoring step uses; an earlier uncheckpointed run agreed to within ± 0.003 on every metric. Temperatures are fitted on validation only. The near-null GPM–BT difference is reported, not used as a gate: this dataset contains no observed triangles and cannot identify a cyclicity advantage.

model	objective	per-seed (mean of 3)			T	calibrated 3-seed ensemble				
		NLL	bal. acc	AUC		NLL	bal. acc	AUC	ECE	ens. sd
Anti-symmetric GPM	helpfulness	0.5543	0.7218	0.7999	1.282	0.5512	0.7257	0.8034	0.1228	0.0293
Anti-symmetric GPM	harmless.	0.5180	0.7556	0.8464	1.297	0.5131	0.7573	0.8512	0.1782	0.0357
Scalar BT	helpfulness	0.5560	0.7214	0.7974	1.270	0.5527	0.7250	0.8009	0.1220	0.0300
Scalar BT	harmless.	0.5212	0.7507	0.8431	1.265	0.5163	0.7544	0.8484	0.1779	0.0357

reward, and Nash versus utilitarian versus Kalai–Smorodinsky aggregation. Hyperparameters and operating points are selected on validation data before final seeds.

E SOLVER AND NEURAL-REALIZATION DIAGNOSTICS

The direct inner solve is exact only for the declared finite response pool and numerical tolerance. It does not make the preference model, finite-pool approximation, or neural projection exact. Final claims require both solver residuals and held-out regression diagnostics.

F EVALUATION PROTOCOL

Natural objective evaluation. The preferred headline evaluation is a blinded human comparison against the common reference, with helpfulness and harmlessness collected as separate tasks. Method identity is hidden, response order and opaque identifiers are randomized independently, ties are allowed, and individual votes are retained. Automatic Beaver reward/cost scores are reported only as in-domain diagnostics because those evaluators are trained from SafeRLHF-family supervision. HarmBench, IFEval, TruthfulQA, Arena-Hard, and AlpacaEval provide complementary safety and capability checks.

G PROMPTED-JUDGE AUDIT

A prompted Qwen3-32B route was audited before large-scale labeling and rejected. Increasing the decoding budget removed the original 4–21% truncation-induced invalid rate, but it did not repair semantic instability. Under the best deliberative protocol, confident swap consistency remained 0.412–0.708 against a predeclared 0.85 gate, and order split-half Spearman correlation remained 0.189–0.428 against 0.75. A factorial audit showed strong lexical A/B bias; opaque identifiers reduced that component but transferred comparable bias to physical position and still failed the gates. No prompted-judge bank, reward model, or policy checkpoint from these labels enters the main experiments.

Table 10: Generated-pool geometry pilot on 200 SAFERLHF validation prompts. “Model conflict” is the frozen preference-model ensemble’s objective disagreement rate on *generated* responses and is not the human-label conflict rate of Table 8. “Pred. cycles” are *model-predicted* closed triangles on the comparator-versus-comparator tournament, where rows and columns index the same response set; no human triangle is observable in this data. Split-half correlations resample the oracle ensemble, not the pool. Selection used validation stability and compute only.

geom.	dup. rate	entropy	ens. sd	GPM/BT sign	model conflict	pred. cycles	split-half r	ρ	sign agr.	dual evals	solve (s)
4+4	0.0073	0.6117	0.0429	0.9102	7.67%	0/800	0.9172	0.9132	0.8667	42	24.8
8+8	0.0167	0.6120	0.0414	0.9061	7.80%	10/11200	0.9242	0.9180	0.8925	30	18.4

Table 11: Final SafeRLHF policy-training configuration. Report the frozen values actually used by all final seeds.

setting	value	setting	value
Reference / learner	Llama-3.1-8B-Instruct	adaptation	full fine-tuning
Policy prompts train/dev/test	7,000/500/1,000	final seeds	42, 43, 44
Pool geometry	—	pair proposal	all unordered learner pairs
Preference models	3-seed GPM / BT ensembles	inner solver	direct finite-pool concave solve
Opponent temperature β	—	proximal coefficient η	—
Policy learning rate	—	optimizer updates	—
Optimizer / scheduler	Adafactor / cosine	warmup / weight decay	0.10/0
Global batch / max length	32/2048	precision / grad. clip	bf16 / 1.0
Dual tolerance / box	—	neural target mode	finite-pool log-ratio
Code / split / config hashes	—		

Table 12: Baseline contract. All matched controls use the same finite pool and neural realization.

method	within-objective representation	aggregation	changed component relative to NBPO
NBPO	adaptive regularized game	Nash	—
Fixed-reference Nash	fixed reference margin	Nash	removes adaptive opponent
BT-RM-Nash	scalar BT reward	Nash	imposes scalar reward representation
Game-utilitarian	adaptive regularized game	equal-weight sum	replaces Nash aggregation
Game-KS	adaptive regularized game	normalized proportional gains	alternative bargaining rule
BT-RM-utilitarian	scalar BT reward	equal-weight sum	scalar representation and aggregation control

Table 13: Neural realization of the finite-pool target. The zero-predictor MSE is $\mathbb{E}[(\hat{h}_t^*)^2]$ on the same split.

row	target RMS	p_{10}	p_{90}	$\ \lambda\ _1$	proj. KKT	inner resid.	identity	norm. MSE	sign agr.	Pearson
NBPO	2.273	−3.841	+1.092	13.71	4×10^{-15}	3×10^{-13}	2×10^{-15}	—	—	—
Fixed-ref. Nash	2.333	−3.959	+1.093	13.87	0	0	2×10^{-15}	—	—	—
BT-RM-Nash	2.473	−4.223	+1.059	2.88	0	0	2×10^{-15}	—	—	—
Game-util.	2.273	−3.843	+1.092	13.71	—	3×10^{-9}	2×10^{-15}	—	—	—
Game-KS	2.276	−3.848	+1.091	13.71	—	7×10^{-9}	2×10^{-15}	—	—	—

Table 14: Direct finite-pool solver scaling. The 7,000-prompt row must be measured rather than extrapolated.

prompts	pool	workers	solve (s)	dual evals	s/eval	proj. KKT	inner resid.	peak RSS (MB)	artifact (s)
200	4+4	32	1.5	31	0.047	3×10^{-14}	3×10^{-11}	467	0.01
200	8+8	32	1.7	35	0.049	4×10^{-15}	5×10^{-12}	498	0.02
200	16+16	32	1.8	32	0.055	2×10^{-14}	10^{-11}	499	0.02
1000	4+4	96	4.4	33	0.132	7×10^{-15}	2×10^{-11}	499	0.03
1000	8+8	96	3.2	26	0.122	4×10^{-15}	6×10^{-12}	504	0.04
1000	16+16	96	4.4	37	0.119	3×10^{-13}	10×10^{-11}	534	0.07
7000	4+4	96	10.6	28	0.379	7×10^{-10}	4×10^{-11}	534	0.11
7000	8+8	96	10.9	27	0.404	2×10^{-15}	3×10^{-11}	544	0.22
7000	16+16	96	15.3	30	0.508	10^{-14}	7×10^{-11}	633	0.39

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Table 15: Blinded human-evaluation record. Fill from the released annotation manifest and imported labels.

item	final value
Prompts / methods / objectives	—
Annotators per pair / tie policy	—
Order and identifier randomization	independent; exact manifest hash —
Repeated quality-control fraction	—
Parse / completion rate	—
Inter-annotator agreement	—
Repeated-item consistency	—
Triangle coverage / observed cycle rate	—

Table 16: Summary of the retired prompted-judge route. Known-answer controls diagnose rubric understanding; natural-pair stability determines admissibility.

protocol	unresolved invalid	min control accuracy	natural-pair stability
P2-long, 3 templates	0.000	0.960	failed swap/order gates
P2-balanced	0.000	0.800	failed controls and stability
Opaque identifiers	0.000	0.940	failed swap/order gates